

Yuanchen Tang

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Education

University of Illinois Urbana-Champaign

Aug 2025 – present

Ph.D. in Computer Science

- GPA: 4.00/4.00

Peking University

Sept 2021 – June 2025

B.S. in Computer Science (Turing Class)

- GPA: 3.80/4.00

Research Interests

My research interests lies at the interface between computer science and economics, particularly in machine learning, algorithmic game theory and reinforcement learning. Recently I am also curious about AI economics.

Papers

The Stability of Data Exchange in Competitive Markets

May 2026

Yuanchen Brian Tang, Jiaxin Song, Bhaskar Ray Chaudhury, Ruta Mehta
openreview.net/forum?id=qQCISzgHHs (*NeurIPS 2026*)

Sealed-bid Auctions on Blockchain with Timed Commitment Outsourcing

Oct 2024

Jichen Li, *Yuanchen Tang*, Jing Chen, Xiaotie Deng
arxiv.org/abs/2410.10607 (IJTCS-FAW 2026)

Research experience

Competitive Data Market

Champaign, USA

UIUC, advised by Prof. Bhaskar Ray Chaudhury and Prof. Ruta Mehta

Oct 2025 – May 2026

- Method: Game theory, Machine Learning
- Key results: We study the stability of data exchange among competing firms. Specifically, can there be a stable data exchange in such competitive environments? We found that in fully rational markets, where customers flock entirely to the topperforming firm, a core-stable exchange always exists for any monotone continuous prediction and utility functions, while in partially rational markets with logit-style demand, core-stable exchanges may fail to exist

Large Aggregative Markov Games

Pittsburgh, USA

Carnegie Mellon University, advised by Prof. Steven Wu

June 2024 – present

- Method: Game Theory, Multi-Agent Planning
- Key results: We developed provably efficient algorithms for multi-agent planning in large general-sum Markov games. This work suggested a new type of Nash-computable games that capture the observation on many real-world large games (congestion control, public goods provision, etc.) that reward functions and dynamics are bounded influenced by unilateral deviation.

Sealed-bid auction on blockchain

Beijing, China

Peking University, advised by Prof. Xiaotie Deng

Dec 2023 – Oct 2024

- Method: Auction theory, Mechanism Design, Cryptography in Blockchain, Incentive Analysis
- Key results: We introduced a time-commitment(puzzle) outsourcing market to address the cost problem of decrypting the puzzle, proposed a fully practical seal-bid auction protocol that incentivized the truthful reporting of bidders' and solvers' private values, and presented the optimal pricing strategy for auctioneers.

Selected Honors and Awards

The Third Prize of Peking University Scholarship	2024
Award for Outstanding Research	2024
John Hopcroft Scholarship in Peking University	2022~2023
Peking University Freshman Scholarship	2021
Chinese Mathematical Olympiad Gold Medal	2020

Skills

Programming: C++, Python, Lean, LaTeX

Sports: Table Tennis(>10 years), Basketball, Baseball

Languages: Chinese(native), English(fluent) TOEFL iBT: 107/120 (Reading 29, Listening 30, Speaking 23, Writing 25(Best 27))